

Crop Weather Calendar of *Rabi* season Soybean: Barisal Region (Districts: Barisal, Patuakhali, Jhalakathi, Barguna & Bhola), Bangladesh

Region: Barisal	Rabi season Soybean								Life span: 110-120 day								
Months	January				February				March					April			
Std. Week/Normal	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Rainfall (mm)	1.3	1.0	1.0	1.7	2.7	3.7	5.0	8.0	5.7	5.3	6.0	14.3	18.0	18.0	21.7	25.0	34.7
Max. Temp. (°C)	25.1	25.2	25.8	26.2	27.0	27.8	28.9	29.7	30.7	31.7	32.7	33.0	32.9	33.3	33.7	33.5	33.5
Min. Temp. (°C)	12.6	12.5	12.7	12.5	14.0	14.6	16.2	16.9	18.3	19.5	20.5	22.1	22.8	23.7	24.0	24.6	24.4
Mean Temp. (°C)	18.9	18.9	19.2	19.4	20.5	21.2	22.5	23.3	24.5	25.6	26.6	27.6	27.9	28.5	28.8	29.1	29.0
RH max (%)	98.0	97.8	97.6	97.6	97.3	97.1	97.0	97.0	96.7	97.1	96.7	96.5	96.5	96.6	96.4	96.0	96.0
RH Min (%)	50.1	49.3	48.5	45.2	46.2	44.8	46.2	45.1	44.9	45.2	45.3	50.7	54.4	58.0	59.8	61.9	62.5
RH mean (%)	74.1	73.6	73.1	71.4	71.7	70.9	71.6	71.0	70.8	71.2	71.0	73.6	75.4	77.3	78.1	78.9	79.3
SS hr (hrs)	46.7	44.0	47.3	52.7	50.3	52.3	51.7	52.0	53.7	54.0	54.3	52.3	50.7	53.0	51.3	54.7	51.7
WD (deg)	209	211	204	216	217	211	214	211	214	226	218	210	198	200	198	197	191
WS(Km/hr)	4.5	4.4	4.5	4.3	5.0	4.6	4.9	4.9	5.1	5.2	5.4	6.3	6.7	6.9	7.0	7.5	7.2
Growth stages																	
	Germination and emergence			Vegetative stage					Reproductive stage					Maturity and Harvesting			
Favourable Environment																	
Temperature	Germination rate lower below 13°C and maximum at 25°C			The optimum temperature is 20°C -30 °C; at 35°C temperatures and above considered inhibitory to production.													
Relative humidity	50% to 70%																
Soil temperature	At least 10°C required for germination and root nodulation; Root zone temperature for adequate nodulation and nitrogen fixation is needed at least 15°C to 17°C whereas 25°C is optimal.																
Sunshine hour	Six to eight hours of sunlight																
Favorable environmental conditions for insects and diseases																	
Hairy caterpillar				Egg, larva and pupa gradually increased with increase in temperature, while total developmental period and adult survival period decreased. The maximum and minimum temperatures for larva development are 39 ± 1°C and 12 ± 1°C.respectively.													
Common Cut warm				Lowest fecundity and fertility of eggs occur at 34.90 °C; the upper development threshold temperature for all stages is 37 °C; The highest fecundity and fertility of eggs occur at 19.55 °C; RH-65% to 85%.													
Leaf roller				Temperatures between 25° C and 30° C are favorable for the survival and multiplication of leaf roller.													
Pod borer				Maximum temperature: 27.48°C -31.44 °C, Minimum temperature: 1 9.73°C -23.91 °C; Average relative humidity ranging from 86.18% - 93.61 %; The weekly average of total rainfall ranging from 11.07-94.33 mm with low or medium rainfall in previous weeks followed by high in current weeks.													
Anthravnose				For disease development maximum temperature ranged from 28 °C to 32 °C and minimum temperature 22 °C to 25 °C, Frequent rainy and warm weather along with prolonged leaf wetness (exceed 12 hrs/day) favored disease development													
Yellow Mosaic Virus	Viral infection and symptom appearance are found most severe at 18 °C to 20 °C but may not be apparent at the temperature higher than 30 °C; Disease development favored by rainfall.																

Weather Warning				
Rainfall	Occurrence of rainfall delayed sowing (5 to 10 mm)	Less than 20 mm		
Cloudy weather	Cloudy condition			
Drought	Too much dry weather creates soil moisture stress			
Duration of staying wet			Soybeans can tolerate brief water logging but long-term causes weathering of seeds.	
Hailstorm			Early rainfall with hailstorms	
Windspeed	15 Km/hr or more			
Temperature	35.5° C or more			

Crop Weather Calendar of *Rabi* season Soybean: Chattogram Region (Districts: Chattogram, Cox’s Bazar, Noakhali, Feni, Lakshmipur), Bangladesh

Region: Chattogram	Rabi season Soybean								Life span: 110-120 day								
Months	January				February				March					April			
Std. Week/Normal	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Rainfall (mm)	0.3	0.7	0.7	1.3	2.0	1.3	2.3	8.3	4.7	5.3	6.3	13.3	21.3	21.3	27.0	34.3	36.0
Max. Temp. (°C)	26.5	26.1	26.6	26.9	27.7	28.7	29.3	29.8	30.6	31.2	32.0	32.0	31.8	32.1	32.6	32.7	32.9
Min. Temp. (°C)	14.1	13.9	14.1	14.1	14.9	15.8	16.9	17.6	18.8	19.8	20.7	21.8	22.7	23.5	24.0	24.4	24.5
Mean Temp. (°C)	20.3	20.0	20.4	20.5	21.3	22.2	23.1	23.7	24.7	25.5	26.3	26.9	27.2	27.8	28.3	28.6	28.7
RH max (%)	93.8	93.1	92.9	91.6	91.0	90.6	90.4	91.0	90.8	91.1	91.3	93.0	93.0	93.3	92.9	92.6	92.8
RH Min (%)	48.2	47.1	44.9	42.3	43.6	41.2	43.3	43.9	44.8	46.6	47.4	54.0	58.2	60.1	60.5	61.6	62.0
RH mean (%)	71.0	70.1	68.9	67.0	67.3	65.9	66.8	67.5	67.8	68.8	69.4	73.5	75.6	76.7	76.7	77.1	77.4
SS hr (hrs)	53.3	50.8	53.8	56.5	56.3	57.8	57.5	58.5	60.3	59.3	56.8	57.0	52.8	56.0	56.0	54.8	55.3
WD (deg)	200	193	203	203	201	203	208	214	221	215	223	218	207	209	205	205	202
WS(Km/hr)	6.6	6.9	6.8	7.0	7.1	7.3	7.7	7.9	8.6	8.7	8.9	9.4	9.7	10.0	11.1	10.6	10.6
																	
Growth stages	Germination and emergence				Vegetative stage				Reproductive stage					Maturity and Harvesting			
Favourable Environment																	
Temperature	Germination rate lower below 13°C and maximum at 25°C				The optimum temperature is 20°C -30 °C; at 35°C temperatures and above considered inhibitory to production												
Relative humidity	50% to 70%																
Soil temperature	At least 10°C required for germination and root nodulation; Root zone temperature for adequate nodulation and nitrogen fixation is needed at least 15°C to 17°C whereas 25°C is optimal.																
Sunshine hour	Six to eight hours of sunlight																
Favorable environmental conditions for insects and diseases																	
Hairy caterpillar					Egg, larva and pupa gradually increased with increase in temperature, while total developmental period and adult survival period decreased. The maximum and minimum temperatures for larva development are 39 ± 1°C and 12 ± 1°C.respectively.												
Common Cut warm					Lowest fecundity and fertility of eggs occur at 34.90 °C; the upper development threshold temperature for all stages is 37 °C; The highest fecundity and fertility of eggs occur at 19.55 °C; RH-65% to 85%.												
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Weather Warning				
Rainfall	Occurrence of rainfall delayed sowing (5 to 10 mm)	Less than 20 mm		
Cloudy weather	Cloudy condition			
Drought	Too much dry weather creates moisture stress			
Duration of staying wet			Soybeans can tolerate brief water logging but long-term causes weathering of seeds.	
Hailstorm			Early rainfall with hailstorms	
Windspeed	15 Km/hr or more			
Temperature	35.5° C or more			

Crop Weather Calendar of *Rabi* season Soybean: Rangpur Region (Districts: Rangpur, Lalmonirhat, Nilphamari, Kurigram, Gaibandha), Bangladesh

Region: Rangpur	Rabi season Soybean												Life span: 110-120 day				
Months	January				February				March					April			
Std. Week/Normal	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Rainfall (mm)	1.0	2.5	2.0	1.0	1.0	1.5	3.5	1.5	2.5	1.0	2.5	5.5	18.5	11.0	19.5	28.5	31.5
Max. Temp. (°C)	22.7	22.4	23.2	23.4	24.7	25.9	26.7	27.5	28.5	29.9	30.8	31.6	31.4	32.1	32.5	32.3	32.0
Min. Temp. (°C)	10.9	10.4	10.8	10.8	11.8	12.7	13.7	14.4	15.3	16.1	17.3	18.4	19.5	20.3	21.1	21.9	21.9
Mean Temp. (°C)	16.8	16.4	17.0	17.0	18.3	19.3	20.2	21.0	21.9	23.0	24.0	25.0	25.4	26.2	26.8	27.1	27.0
RH max (%)	96.0	96.0	95.8	95.7	95.5	94.3	93.7	92.9	91.4	89.8	88.7	88.1	87.7	88.5	88.4	89.9	90.9
RH Min (%)	58.5	57.8	56.2	54.2	51.6	48.0	47.1	45.3	42.8	39.6	40.8	42.5	48.8	49.7	50.7	56.7	59.0
RH mean (%)	77.3	76.9	76.0	74.9	73.5	71.2	70.4	69.1	67.1	64.7	64.7	65.3	68.3	69.1	69.5	73.3	75.0
SS hr (hrs)	36.0	36.5	39.5	41.0	44.0	48.5	50.0	51.5	56.5	57.5	57.0	54.0	49.0	51.5	51.0	47.5	47.0
WD (deg)	199	214	199	210	208	192	201	201	201	207	196	179	152	148	141	135	118
WS(Km/hr)	5.7	5.7	5.7	5.6	6.2	6.6	7.5	7.2	7.7	7.8	8.2	8.7	8.7	8.3	8.4	8.3	8.6
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